Kopibara RLab Agent: An autonomous ML researcher that searches over executable hypotheses, one measured code change at a time.

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1 Objective and evaluation

The task is to automate the engineering loop for a recommendation ranker: inspect the benchmark contract, construct features, train and tune a model, evaluate on public validation data, and revise the code. KuaiRand ranks logged impressions within each user. The relevance label is `long_view`, and the primary score is the mean of GAUC and nDCG@5. KuaiRand-Pure is the required benchmark; KuaiRand-1K is the larger bonus variant.

The controller develops against train and validation splits only. Test rows are used only to produce the final score file, after their labels and feedback have been masked. The hidden split is never read during search. The organizer's convergence rule is an improvement of more than 0.002 over three consecutive iterations, with 50 iterations and six hours as backstops.

The official Pure Factorization Machine reference is 0.6016 on validation primary. For 1K, the starter kit does not publish an official reference, so the run measures the supplied FM pipeline at 0.6079 and labels it as a runtime reference. This distinction matters: the Pure comparison is against the organizer's published number, while the 1K comparison is only an internal benchmark.

2 Agent design

The controller uses `gpt-5.6-luna` with low reasoning effort as its planner. Its design follows the executable code-search framing of AIDE and the reproducible benchmark discipline represented by MLE-bench. At each turn, the planner receives the benchmark contract, the current solution tree, the best measured node, and the source of the node selected as parent. It returns one testable hypothesis and a small JSON list of exact-match replacements.

The runner applies a replacement only when its anchor occurs exactly once. The planner may edit `history_lgbm.py` only, with at most four replacements, and the candidate must retain the existing command-line interface and validation output. A denylist rejects new network, shell, dynamic-import, or code-evaluation paths. Candidate processes run with a copied environment that removes `OPENAI_API_KEY`, so the language-model credential cannot reach generated code.

Each candidate is compiled and run in a bounded subprocess. The runner parses the candidate's validation metrics, falls back to its `metrics.json` when stdout is incomplete, and stores the hypothesis, exact diff, metrics, token counts, command, and recovery events. The controller keeps the highest-scoring measured node as the next parent. A failed candidate gets one repair request; a second failure is recorded as `recovered_failure` and cannot become a parent.

3 Seed pipeline

The seed is a grouped LightGBM ranker. It combines categorical context fields for the user, video, author, tab, duration bucket, hour, random-exposure flag, date, 15-minute time bucket, weekday, and released extra fields with continuous duration and time features. The ranking groups are users, matching the evaluation unit rather than treating the dataset as an ungrouped binary-classification problem.

The main inductive bias is a chronological history feature builder. For each user, video, author, and user-video pair, it records prior interaction count, feedback rates, log-transformed cumulative feedback, the last-seen timestamp transform, and the log time since the previous observation. All released feedback signals are available to these histories as auxiliary signals, while `long_view` remains the only scored label. The state is updated only after an observed train or validation row, so a row cannot use its own outcome or a later outcome. Test rows receive features but do not update the state.

4 Search results

The selected validation results are summarized in Table 1. The reported 1K primary improves by 0.1419 over its measured FM reference; it is not an official hidden-test delta. The Pure primary improves by 0.0283 over the organizer's official validation reference.

Benchmark	Reference	GAUC	nDCG@5	Primary	Delta
KuaiRand-Pure	0.6016 official	0.7059	0.5538	**0.6299**	**+0.0283**
KuaiRand-1K	0.6079 measured	0.7107	0.7888	**0.7498**	**+0.1419**

Table 1: Best validation results. The 1K reference is measured locally, not organizer-published.

The Pure seed was already the best measured node at 0.6299. The search then checked three children: a planned XE-NDCG change that was not effective because the fixed Pure command still passed `--objective lambdarank`, truncation 10, and disabled query normalization. Their primary scores were 0.6299, 0.6264, and 0.6268 respectively. The no-op branch is retained in the run log as an execution mismatch rather than being presented as evidence for or against XE-NDCG.

The 1K run found a much larger improvement. The seed scored 0.6038, below the measured FM reference. Enabling grouped `rank_xendcg` raised primary to 0.6920, with nDCG@5 increasing to 0.6971. Increasing `num_leaves` from 31 to 63 then raised GAUC to 0.7034 and nDCG@5 to 0.7588, for primary 0.7311. A latest-feedback feature extension, smaller leaves, path smoothing, and truncation levels 10 and 3 were all tested from that parent and rejected.

The next accepted change added `lambda_l2=2.0`, moving primary to 0.7487. The planner then tested stronger, weaker, and intermediate L2, more boosting rounds, a slower learning rate, L1, feature subsampling, and categorical smoothing. Every one was lower than the L2=2.0 parent or unchanged. Finally, `max_bin=511` improved primary to 0.7493. `max_bin=1023` fell to 0.7473, `max_bin=640` reached 0.7475, and the intermediate `max_bin=767` became the best node at 0.7498.

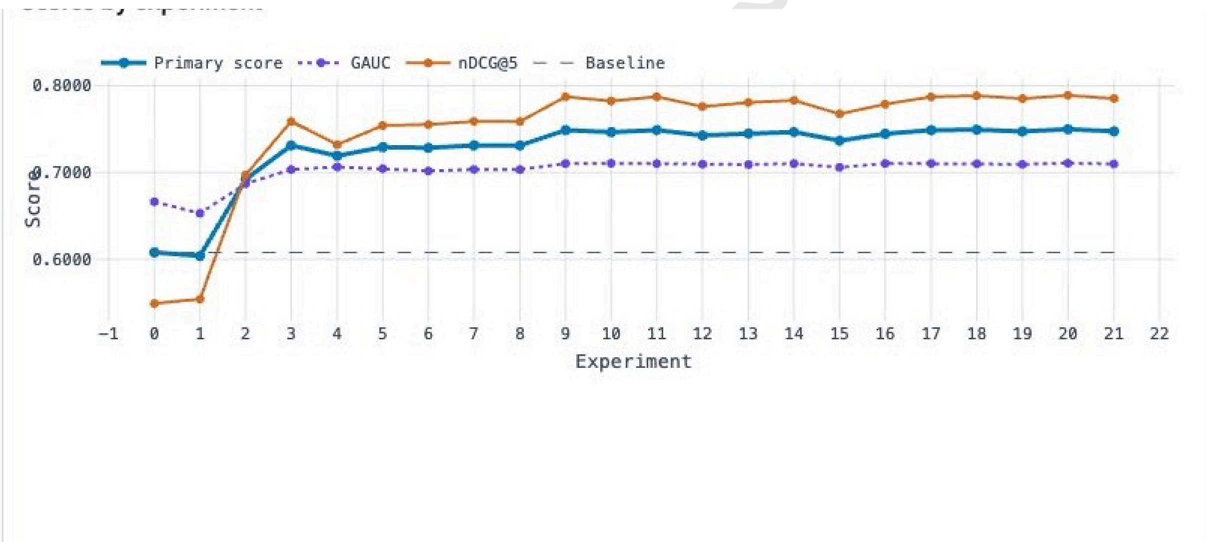

Figure 1: 1K validation trajectory. The lines show primary, GAUC, nDCG@5, and the measured reference.

The trajectory in Figure 1 shows two different regimes. The objective change and tree capacity produced the large gains; once the ranker reached 0.7311, the search became local regularization and quantization tuning. This is the useful behavior of the agent on this task: it moved from a model-family decision to narrow, measured changes, kept only improvements, and preserved the best checkpoint when later trials failed or regressed.

5 Complete decision record

The full 1K sequence is below. The primary column is sufficient to explain parent selection because the controller compares the mean score; the per-metric values remain in each JSON iteration log. “Rejected” means the candidate ran but did not exceed the current best. “Failed” means both the original candidate and its repair timed out.

Iter.	Decision tested	Primary	Outcome
0	Seed: context plus four lagged multi-feedback histories	0.6038	Kept
1	Use grouped `rank_xendcg` instead of LambdaRank	0.6920	Kept
2	Increase `num_leaves` from 31 to 63	0.7311	Kept
3	Add latest feedback at each history scope	0.7192	Rejected
4	Reduce `min_data_in_leaf` to 20	0.7292	Rejected
5	Add `path_smooth=20.0`	0.7285	Rejected
6	Increase ranking truncation to 10	0.7311	Rejected
7	Reduce ranking truncation to 3	0.7311	Rejected
8	Add `lambda_l2=2.0`	0.7487	Kept
9	Increase L2 to 4.0	0.7465	Rejected
10	Allow 600 boosting rounds	0.7487	Rejected
11	Reduce L2 to 1.0	0.7427	Rejected
12	Use learning rate 0.03 and 700 rounds	0.7448	Rejected
13	Use intermediate L2 value 2.5	0.7466	Rejected
14	Add mild L1 regularization	0.7366	Rejected
15	Use feature fraction 0.85	0.7445	Rejected
16	Smooth categorical splits with `cat_smooth=20.0`	0.7487	Rejected
17	Increase `max_bin` to 511	0.7493	Kept
18	Increase `max_bin` further to 1023	0.7473	Rejected
19	Use intermediate `max_bin=767`	0.7498	Kept
20	Tune near the optimum with `max_bin=640`	0.7475	Rejected
21	Reduce XE-NDCG sigmoid scale to 0.5	–	Failed: timeout; retained 019

Table 2: Complete 1K search sequence recorded by the autonomous controller.

The Pure sequence is shorter and is shown separately in Table 3. It reached the organizer convergence rule after three children. The first child demonstrates why the run logs matter: its planner hypothesis was sensible, but the actual fixed command overrode the candidate's new default. The measured record makes that limitation visible and keeps the final claim narrow.

Iter.	Decision tested	Primary	Outcome
0	Seed: grouped LightGBM with leakage-safe histories	0.6299	Kept
1	XE-NDCG support planned; command still used LambdaRank	0.6299	No-op; rejected
2	Increase ranking truncation to 10	0.6264	Rejected
3	Disable query-size lambda normalization	0.6268	Rejected

Table 3: Pure search sequence. Scores are validation primary.

6 Reproducibility, safety, and accounting

The run artifacts preserve each hypothesis, exact code diff, validation metrics, subprocess command, token count, and recovery event. The final row-aligned CSV passed the starter-kit's submission checker: 170,588 rows for Pure and 4,132,081 rows for 1K. Both manifests record zero manual interventions and `hidden_test_access: false`.

Benchmark	Run ID	Iters.	Wall-clock	LLM tokens	Manual	Stop
KuaiRand-Pure	20260831T111545Z	3	5m 5s	12,901	0	Converged
KuaiRand-1K	20260831T124656Z	21	6h 7m	115,212	0	Wall-clock cap

Table 4: Resource and autonomy accounting from the final manifests.

The 1K run also records one planner recovery after a response exceeded the four-edit limit. At iteration 21, the agent proposed reducing the XE-NDCG sigmoid scale. The candidate and its repair both timed out after 375.92 seconds, so the controller marked the branch `recovered_failure`, retained node `019-child`, and did not replace the submission checkpoint. This is a failure-handling event, not a hidden-test result.

The reported metrics are public-validation results. Hidden-test scores are unavailable until organizer evaluation, so no hidden-test improvement is inferred. The Pure result is comparable with the official reference; the 1K result is reported against the measured runtime reference only. The source implementation, run logs, manifests, models, scores, and submissions are included in the project archive.